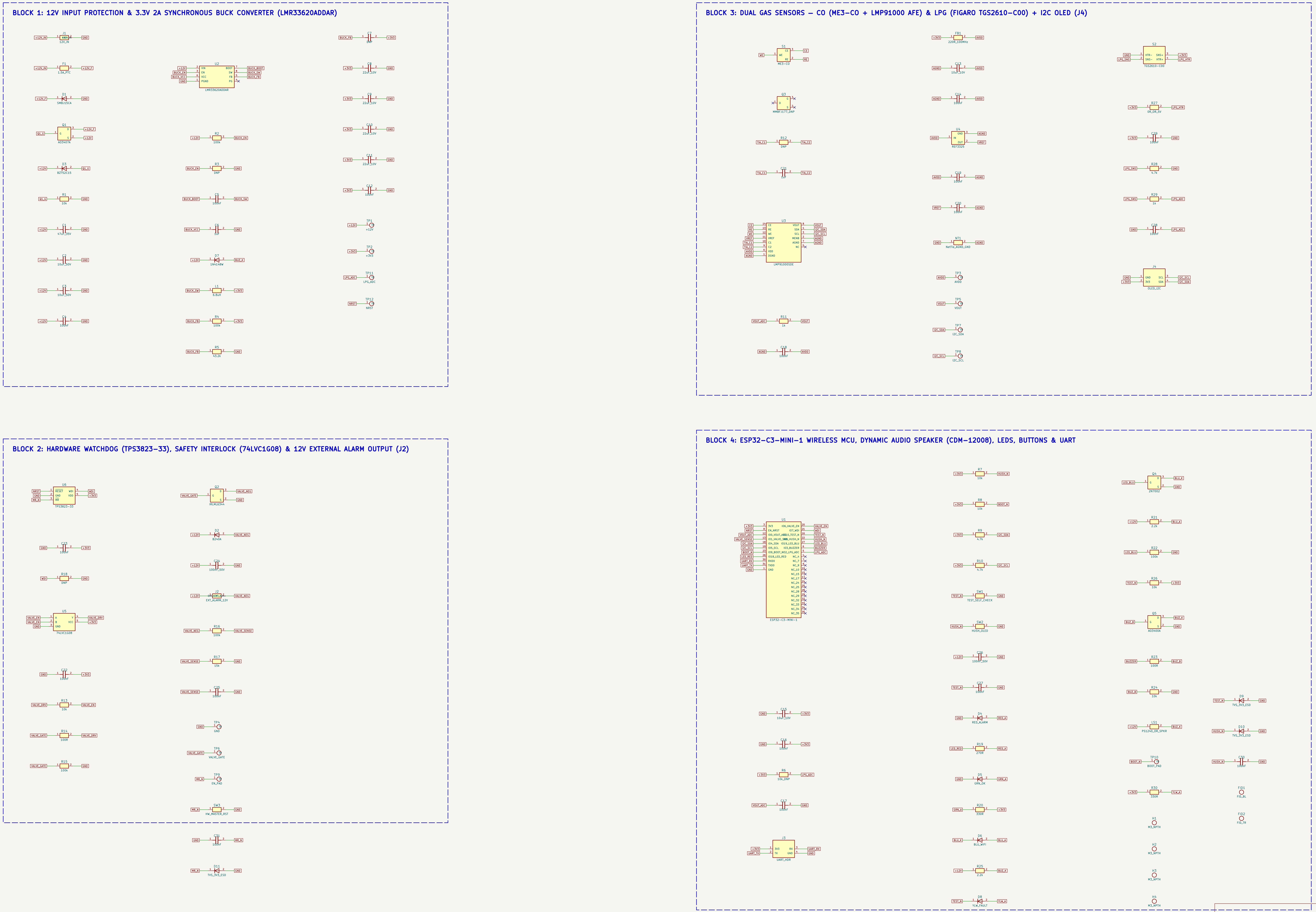
BLOCK 1: 12V INPUT PROTECTION & 3.3V 2A SYNCHRONOUS BUCK CONVERTER (LMR33620ADDAR)
BLOCK 2: HARDWARE WATCHDOG (TPS3823-33), SAFETY INTERLOCK (74LVC1G08) & 12V EXTERNAL ALARM OUTPUT (J2)
BLOCK 3: DUAL GAS SENSORS — CO (ME3-CO + LMP91000 AFE) & LPG (FIGARO TGS2610-C00) + I2C OLED (J4)
BLOCK 4: ESP32-C3-MINI-1 WIRELESS MCU, DYNAMIC AUDIO SPEAKER (CDM-12008), LEDS, BUTTONS & UART
U2
LMR33620ADDAR
U6
TPS3823-33
U5
74LVC1G08
Q2
IRLML6344
J2
EXT_ALARM_12V
U3
LMP91000SDE
U4
REF3325
S1
ME3-CO
S2
TGS2610-C00
J4
OLED_I2C
U1
ESP32-C3-MINI-1
J3
UART_HDR
LS1
PS1240_DR_SPKR

Power: 12V Protected Input + LMR33620ADDAR 3.3V Buck | Safety: TPS3823-33 + 74LVC1G08 + IRLM
MCU: ESP32-C3-MINI-1 | AFE: LMP91000SDE + REF3325 + Winsen ME3-CO
Antigravity Hardware Engineering
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Title: CO Detector & Automatic 12V Gas Shut-Off Valve Controller

Size: A1 | Date: 2026-11-07 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 1/1